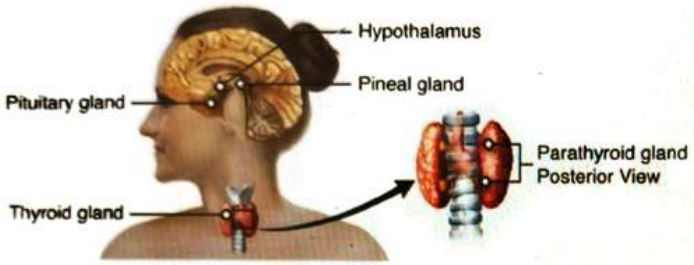
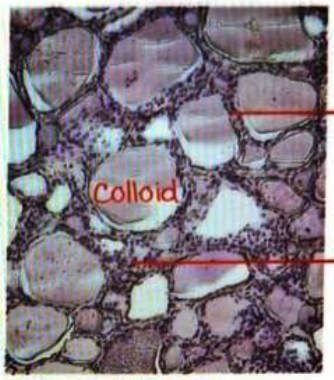
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THYROID GLAND

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Location



Follicular cells
(Produce T3, T4)

Parafollicular C cells
(Produce calcitonin)

Histology

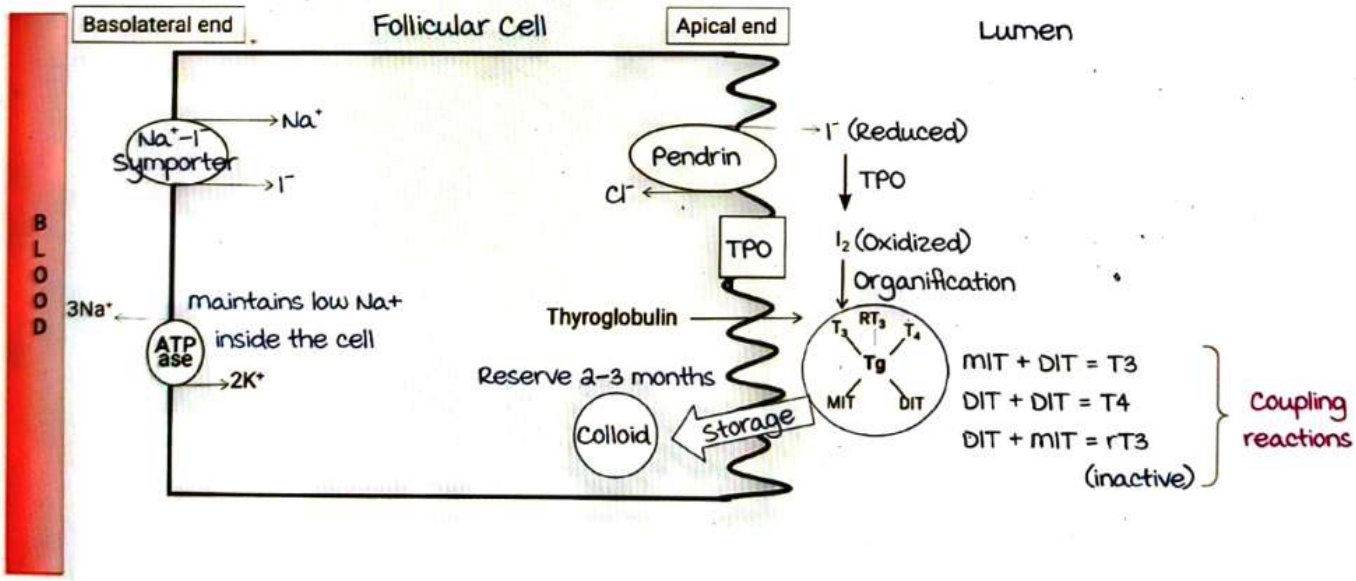
Thyroid Hormone

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SYNTHESIS :

Raw materials :

- Tyrosine : From thyroglobulin (Tg).
- Iodine : Present as Iodide in diet.



Mechanism :

1. Sodium Iodide Symporter (NIS) :

Location : Basolateral end of follicular cells.

Function : Iodide trapping.

Type : 2° active transport.

Other locations : Salivary gland, mammary gland, placenta.

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Wolf - Chaikoff effect :

- Concept : Excess iodide through NIS → Inhibits organification → ↓ T3, T4.
- use : Rx with lugol's iodine prior to thyroid Sx.

2. Iodide - chloride antiporter/pendrin :

Locations : Thyroid gland, inner ear.

Mutation :

- Goiter.
- Sensorineural hearing loss.

} Pendred syndrome

3. Thyroid Peroxidase (TPO) : ~~TPO~~ →

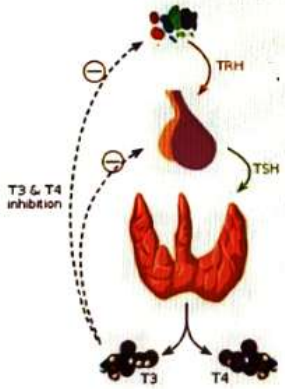
Function : Reduced iodide ~~→~~ oxidised I_a → T3, T4.

Anti-thyroid drugs :

Propylthiouracil, carbimazole, methimazole

Regulation :

Negative feedback control system : Actions of thyroid stimulating hormone (TSH) :



↑ vascularity of gland ↑ iodide trapping

↑ T3, T4 synthesis

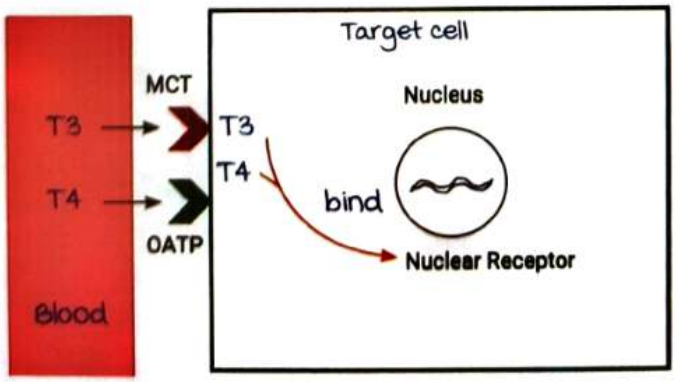
BINDING PROTEINS :

Colloid breakdown → T3, T4 → Released into blood.

99% bound to binding proteins — 1% Free (Responsible for actions)

- Albumin
- Thyroid binding globulin (TBG) : Highest affinity
- Pre Albumin/transthyretin : Binds T3, T4.

RECEPTOR :



mono carboxylase transporter (MCT)

Organic anion transporting polypeptide (OATP)

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DIFFERENCES B/W TRIIODOTHYRONINE (T3) & THYROXINE (T4):

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	T3	T4
Potency	most potent	Least potent
Half-life	1 day	7 days
Plasma protein binding	Less	Tight binding
Action	Rapid	Long acting
Quantity in colloid	Less	more
Form	Physiological: metabolic actions	Pharmacological: Drug form

PERIPHERAL CONVERSION OF T4 → T3:

T4 —Deiodinase→ T3.

Deiodinase: Contains 21st amino acid, selenocystine.

ACTIONS:

Mediated by T3.

1. Increases Basal Metabolic Rate (BMR):

↑ metabolism even at rest → ↑O2 consumption throughout the body except:

- Adult brain.
- Uterus.
- Testes.
- Spleen.

↑ metabolic end products

↓

Vasodilation

↓

↓ Peripheral Vascular resistance

2. Physiologic Uncouplers:

T3 —Stimulates→ Uncoupling protein 1 (UCP1) → Heat dissipation → Thermogenic action (↑ Heat)

Hyperthyroidism → ↑↑ heat generation → Heat intolerance.

Hypothyroidism → No heat generation → Cold intolerance.

3. Catabolism:

Carbohydrate metabolism: Glycogenolysis → ↑ blood glucose.

Lipid metabolism: Lipolysis → ↑ free fatty acids.

Protein metabolism: Proteolysis.

Cholesterol metabolism: ↓ blood cholesterol.

Hypothyroidism → ↑ Blood cholesterol → Atherosclerosis & heart disease.

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4. Effect on Glycosaminoglycans (GAGs):

- T3 prevents accumulation of GAGs (Hyaluronic acid, chondroitin sulfate).
- Hypothyroidism → Accumulation of GAGs → Swelling up - myxedema & combining with water (Puffiness of face).

5. During development:

T3 → Brain growth (↑ myelination in CNS)
T3 → Bone growth

Congenital hypothyroidism:

- mental retardation (m/c reversible cause): Low IQ.
- Stunted growth.

6. On heart:

↑ responsiveness of heart of circulating catecholamines (Norepinephrine).

↓

↑ HR
↑ Stroke volume
↑ Cardiac contractility
— β-receptor mediated

Thyroid Storm: ↑ T3 → ↑ action of NE on heart.
Rx: β-blocker (Propranolol).

7. On lungs:

T3 → ↑ metabolism → ↑ O_a consumption → Hypoxia → Shift of O_a dissociation curve (ODC) to right
↓
Stimulates respiration
↓
↑ respiratory rate.

8. On GIT:

↑ GIT motility & ↑ GI secretions.
Hypothyroidism → ↓ T3 → ↓ motility → Constipation.

9. On reproductive system:

Permissive role in normal reproductive function.
In liver, T3 ↑ sex steroid binding globulin.

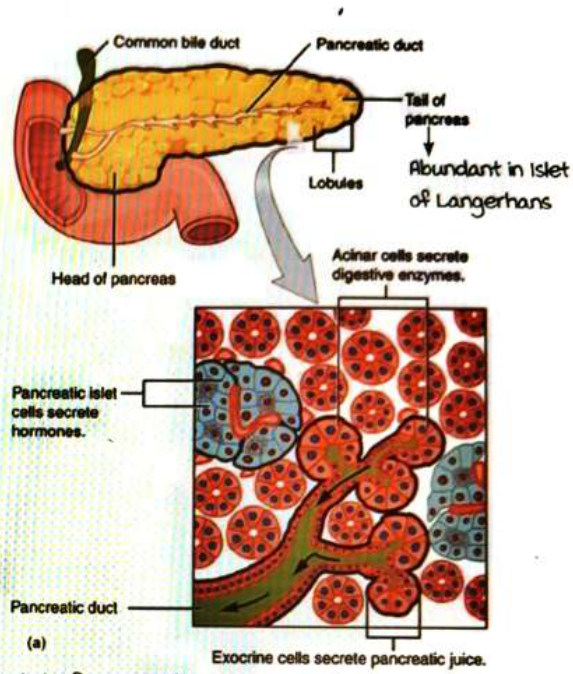
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ENDOCRINE PANCREAS

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Cell type	Hormone(s) secreted & functions
α cells	• Glucagon : ↑ blood glucose.
β cells (majority of cells : 55 %)	• Insulin : ↓ blood glucose. • Amylin : - Potentiates action of insulin. - Drug form : Pramlintide (Rx of DM). • C peptide (Endogenous insulin marker) : - C peptide assay used to estimate insulin release. (C peptide : Insulin = 1:1 → Equimolar ratio)
δ cells	• Somatostatin : Universal inhibitor.
F cells (Least % of cells)	• Pancreatic polypeptide (PP).



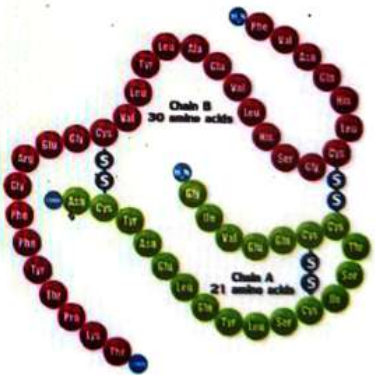
Islet of Langerhans aka Endocrine pancreas

Insulin

00:07:35

Properties :

- Large protein hormone.
- Contains 51 amino acids.
- First protein :
  - Completely sequenced (By Sanger).
  - Produced by recombinant DNA technology.
- Zinc stabilizes structure of insulin.
- Mutation in HNF (Hepatocyte Nuclear Factor) → Maturity Onset Diabetes of the Young (MODY).



Structure

Release :

Food intake → ↑ Blood glucose → Insulin release → ↓ Blood glucose

Negative feedback system

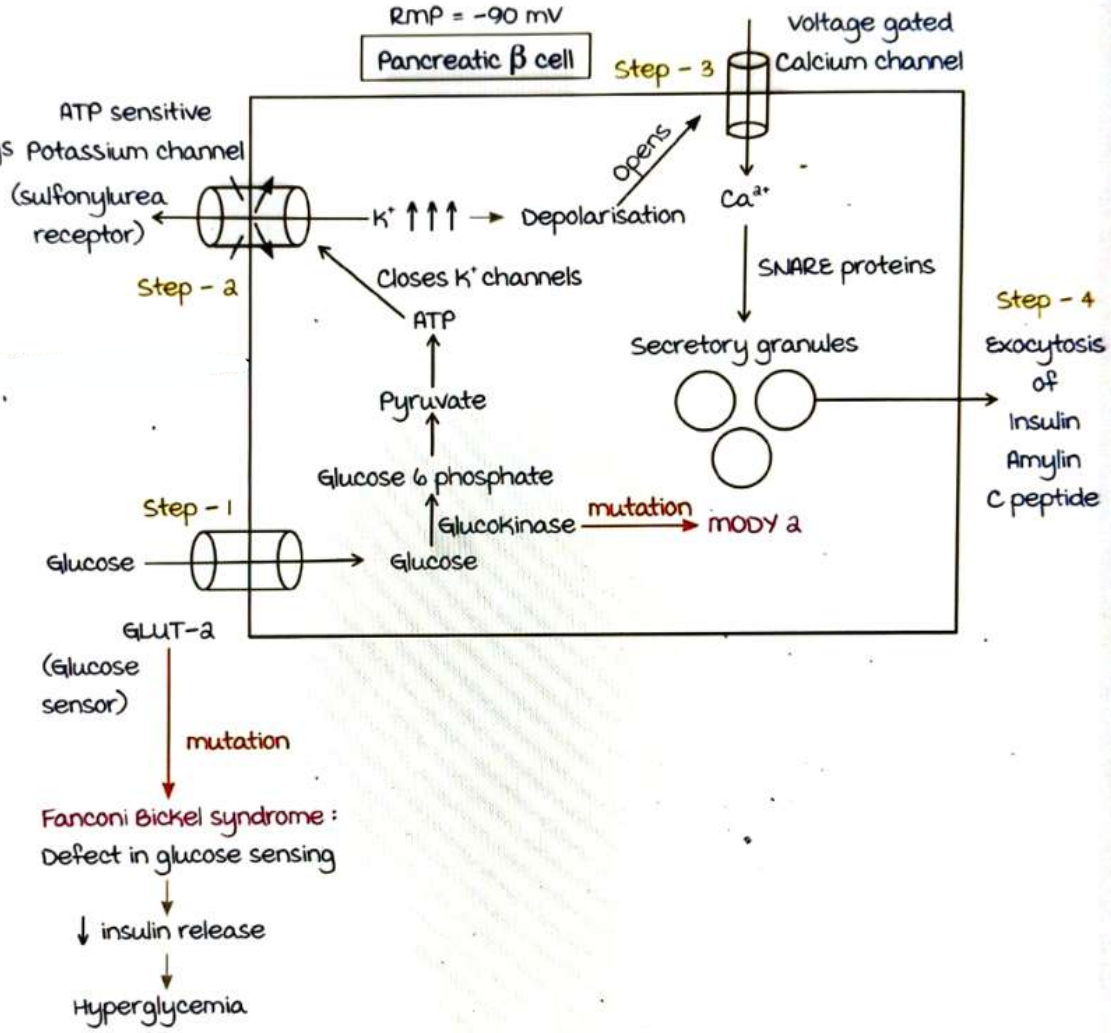
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↓ Insulin ← Opened
used in Rx of insulinoma.
Eg : Diazoxide

↑ Insulin ← Closed
used in Rx of DM.
Eg : Sulfonylureas (Insulin secretagogues)

Drugs



Transport :

- Sites (Insulin dependent areas) : muscle, adipose tissue, heart.
- Process : Facilitated diffusion.
- Transport protein : GLUT-4.
- During exercise, glucose enters skeletal muscle independent of insulin.

Exercise —Activates→ AMP ——→ Insertion of GLUT-4 in skeletal muscle

AMP-activated Kinase pathway

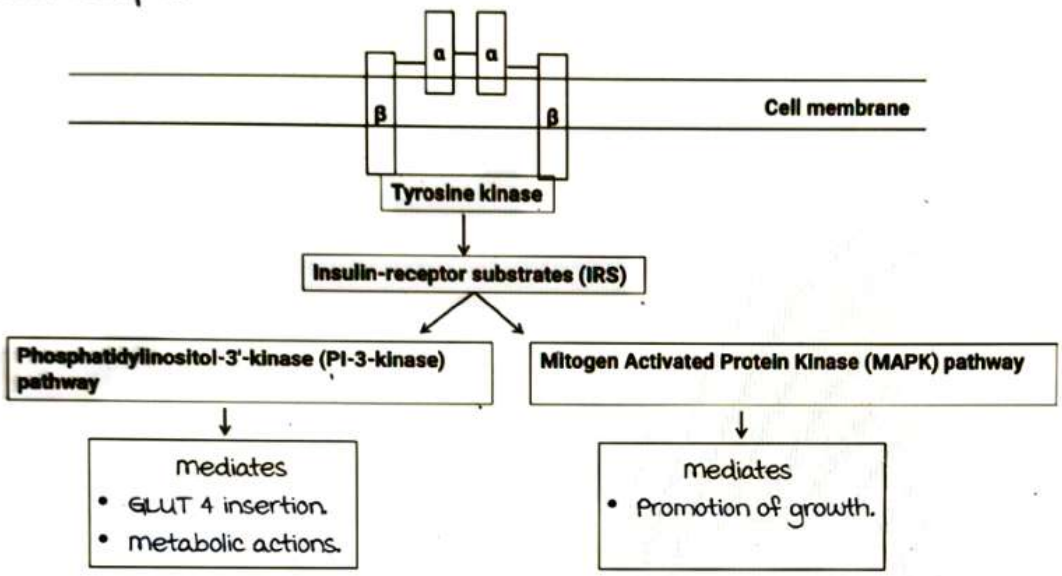
- Significance : Exercise is important in Rx of DM.

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Insulin receptor :

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Actions :

Time dependent

Rapid actions (within seconds)
- ↓ Blood glucose (via GLUT-4).
- ↓ K+ levels.
  - Rx of hyperkalemia : Insulin + dextrose.

Long term actions (in days)
Major anabolic hormone : Storage.

Pathways mediated

Pathways stimulated :
- Glycogenesis.
- Lipogenesis.
- Glycolysis.

Pathways inhibited :
- Glycogenolysis.
- Neoglucogenesis.
- Lipolysis.
- Ketogenesis.

Factors affecting insulin secretion :

↑ Insulin Secretion	↓ Insulin Secretion
1. High blood glucose. 2. α cells : Glucagon. 3. Amino acids : Arginine, Leucine. 4. Autonomic nervous system : Parasympathetic : Vagal stimulation, acetylcholine. Sympathetic : Only β receptor stimulation. 5. Gastrin, CCK, secretin. 6. Incretins. 7. Protein rich meal : Also ↑ glucagon to prevent hypoglycemia.	1. Somatostatin. 2. Streptozocin, Alloxan : Destroy β cells. Used only for research experiments. 3. Sympathetic : α action (Dominant action).

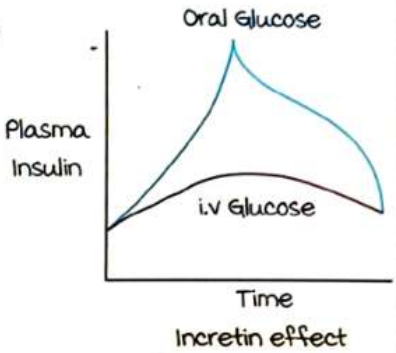
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Applied aspect of incretins :

- Glucagon like Peptide I (GLP-I), Glucose dependent insulinotropic peptide (GIP).
- Degraded by Dipeptidyl peptidase 4 (DPP-4).
- Oral glucose → Stimulate incretins (GIT hormones)
  → ↑ insulin release compared to IV glucose.



Incretin effect

Management of diabetes :

- GLP-I analogues : Exenatide.
- Dipeptidyl peptidase 4 (DPP-4) blockers : Gliptins (Eg. sitagliptin).
  - ↓ degradation → Prolonged action of GLP-I.

Glucose homeostasis :

Decrease in blood glucose
(Hypoglycemic hormone) : Insulin.

Blood glucose < 30 mg/dl → Coma.

Increase in blood glucose
(Counter regulatory hyperglycemic hormones) :

- Growth hormone.
  - Dawn phenomenon : Early morning ↑ in glucose.
- Glucagon.
- T_3.
- Cortisol.
- Epinephrine.

Feedback